SHEET 2

1- Write a program that prints all solutions to the quadratic equation . (aX^2 + bX + C = 0) . Read in a, b, c and use the quadratic formula. If the discriminant is (b^2 – 4ac) negative, display a message stating that there are no solutions. Else solve it .

2- A year with 366 days is called a leap year. A year is a leap year if it is divisible by four (for example, 1980), except that it is not a leap year if it is divisible by 100 (for example, 1900); however, it is a leap year if it is divisible by 400 (for example, 2000). There were no exceptions before the introduction of the Gregorian calendar on October 15, 1582 (1500 was a leap year). Write a program that asks the user for a year and computes whether that year is a leap year. Your program should contain a single if statement.

3-Write a program that asks the user to enter a month (1 for January, 2 for February, and so on) and then prints the number of days in the month. For February, print "28 or 29 days".

example:

Enter a month: 5

30 days.

4- :Assume we want to implement a football game. A player can do the following:

- ***Press "w" Move up***
- ***Press "s" Move down***
- ***Press "d" Move right***
- ***Press "a" Move left***
- ***Press "k" or "j" Kick the ball***
- ***Press "p" Pass the ball***
- ***Press "o" Over the ball***

This should be case insensitive so when user inputs "w" or "W", it should move up. This should apply for all entered letters; both upper and lower case characters .should work

Write a program that reads an input character and prints the proper movement .depending on the input character

.Hint: It should be implemented using switch case statement

5- Write a C++ program that designs a calculator .

The program should read from the user two integer numbers and a character which indicates the type of operation desired.

Hint : solve it by switch

****6- How can you find the minimum/maximum of three numbers using the conditional operator ,in just one line ?***

Hint : Google for ternary operator in C++

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6. Complete the following truth table by finding the truth values of the Boolean expressions for all combinations of the Boolean inputs p, q, and r.

p	q	r	(p && q) || !r	!(p && (q || !r))
0	0	0		
0	0	1		
0	1	0		